

Exploratory Data Analysis (EDA)

Summary Report

1. Introduction

Credit risk assessment is an essential process for financial institutions to evaluate whether customers are likely to repay their loans or become delinquent. This project performs an Exploratory Data Analysis (EDA) on a customer credit dataset to understand customer characteristics, identify data quality issues, and discover the financial and behavioral factors that influence delinquency. The analysis provides valuable insights that can support the development of machine learning models for predicting high-risk customers and improving lending decisions.

2. Dataset Overview

The dataset consists of **500 customer records** with **19 features** describing customer demographics, financial information, credit behavior, and repayment history.

Features Included

- Customer_ID
- Age
- Income
- Credit_Score
- Credit_Utilization
- Missed_Payments
- Delinquent_Account
- Loan_Balance
- Debt_to_Income_Ratio
- Employment_Status
- Account_Tenure
- Credit_Card_Type
- Location
- Month_1 to Month_6 (Monthly Payment Status)

The dataset contains a combination of **numerical** and **categorical** variables, making it suitable for both statistical analysis and predictive modeling.

3. Missing Data Analysis

The dataset intentionally contains missing values to simulate real-world financial data.

Feature	Missing Values
Income	40
Loan_Balance	30
Credit_Score	3

Observations

- Approximately **8% of Income values** are missing.
- **Loan Balance** contains moderate missing values.
- **Credit Score** has very few missing records.
- All remaining variables are complete.

Recommended Treatment

- Replace missing numerical values using the **median**, as it is less affected by extreme values.
 - Verify missing records before training machine learning models to avoid biased predictions.
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4. Key Findings and Risk Indicators

The exploratory analysis identified several important factors related to customer credit risk.

Key Findings

- Most customers belong to the working-age population (18–75 years).
- Customer income is distributed across a wide range without extreme outliers.
- Credit scores vary from poor to excellent, providing useful information for risk assessment.
- Loan balances differ significantly among customers, reflecting different debt burdens.
- Monthly payment behavior (On-time, Late, Missed) provides valuable information about repayment patterns.
- Employment status contains inconsistent labels (such as "EMP", "Employed", and "employed"), which should be standardized before modeling.
- The correlation matrix shows **weak linear relationships** between numerical variables, indicating that delinquency is likely influenced by multiple factors rather than a single variable.

High-Risk Indicators

Indicator	Importance
Low Credit Score	Indicates poor credit history and a higher probability of loan default.
High Credit Utilization	Suggests heavy dependence on available credit, increasing repayment risk.
High Loan Balance	Large outstanding loans increase financial pressure on customers.
High Debt-to-Income Ratio	Higher debt relative to income reduces repayment capacity.
Multiple Missed Payments	Past payment behavior is a strong predictor of future delinquency.
Existing Delinquent Account	Customers already marked as delinquent are more likely to default again.
Unstable Employment	Irregular income may reduce repayment ability.
Repeated Late or Missed Monthly Payments	Consistent payment delays indicate higher financial risk.
Short Account Tenure	Limited credit history makes customer behavior less predictable.

5. AI & GenAI Usage

Artificial Intelligence (AI) and Generative AI (GenAI) were used to improve the efficiency and quality of the analysis.

AI Applications

- Assisted in identifying missing values and data quality issues.
- Supported statistical analysis and feature interpretation.
- Helped identify important financial risk indicators.
- Assisted in understanding relationships between customer attributes.

Generative AI Applications

- Generated Python code for data cleaning and visualization.
- Assisted in creating histograms, count plots, scatter plots, box plots, and correlation heatmaps.
- Helped summarize EDA findings into clear and understandable reports.
- Assisted in documenting project observations, conclusions, and recommendations.

The use of AI and GenAI reduced manual effort, improved productivity, and enhanced the overall quality of the data analysis process.

6. Conclusion & Next Steps

The exploratory data analysis indicates that the dataset provides useful information for understanding customer credit behavior and predicting delinquency. Although the dataset contains a small number of missing values, they can be effectively handled through appropriate preprocessing techniques. Financial variables such as **Credit Score, Credit Utilization, Loan Balance, Debt-to-Income Ratio, Missed Payments, and Monthly Payment History** appear to be the most important indicators of customer credit risk.

Next Steps

- Handle missing values using suitable imputation techniques.
- Standardize categorical variables (e.g., employment status labels).
- Encode categorical features for machine learning.
- Perform feature engineering to improve predictive performance.
- Train and compare classification models such as **Logistic Regression, Decision Tree, Random Forest, XGBoost, and Gradient Boosting**.
- Evaluate model performance using **Accuracy, Precision, Recall, F1-Score, ROC-AUC, and Confusion Matrix**.
- Deploy the best-performing model to assist financial institutions in identifying high-risk customers and supporting better credit approval decisions.

Overall, the analysis establishes a strong foundation for developing an accurate and reliable **customer credit delinquency prediction system** using machine learning.